

On-demand synthesis of organozinc halides under continuous flow conditions

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Organozinc reagents are versatile building blocks for introducing C(sp²)-C(sp³) and C(sp³)-C(sp³) bonds into organic structures. However, despite their ample synthetic versatility and broad functional group tolerance, the use of organozinc reagents in the laboratory is limited because of their instability, exothermicity and water sensitivity, as well as their labor-intensive preparation. Herein, we describe an on-demand synthesis of these useful reagents under continuous flow conditions, overcoming these primary limitations and supporting widespread adoption of these reagents in synthetic organic chemistry. To exemplify this procedure, a solution of ethyl zincbromoacetate is prepared by flowing ethyl bromoacetate through a column containing metallic zinc. The temperature of the column is controlled by a heating jacket and a thermocouple in close contact with it. Advice on how to perform the procedure using alternative equipment is also given to allow a wider access to the methodology. Here we describe the preparation of 50 ml of solution, which takes 1 h 40 min, although up to 250–300 ml can be prepared with the same column setup at a rate of 30 ml per h. The procedure provides the reagent as a clean solution with reproducible concentration. Organozinc solutions generated in flow can be coupled to a second flow reactor to perform a Reformatsky reaction or can be collected over a flask containing the required reagents for a batch Negishi reaction.

INTRODUCTION

Accessing new chemical spaces is a key element of finding new biologically active products. Organic synthesis plays a fundamental role in drug discovery and development¹. For instance, the development of cross-coupling reactions has opened new avenues in building carbon–carbon bonds and offers access to new chemical entities. However, applications of these methodologies have been focused on C(sp²)-C(sp²) and C(sp)-C(sp²) processes that result in the formation of molecules with linear or disc dispositions². Increasing the accessibility to methodologies that allow the formation of C(sp²)-C(sp³) and C(sp³)-C(sp³) will support more spherical or 3D structures. Increasing the prevalence of sp³ centers in bioactive molecules has been associated with an improvement in the pharmacokinetic properties of such molecules^{3,4}.

Organozinc compounds are highly reactive reagents with proven utility in the synthesis of complex functionalized products^{5,6}. Their most common application is in transition metal–catalyzed cross-coupling reactions (Fig. 1; refs. 7–10). Alkyl zincbromoacetates are also noteworthy for their use in Blaise and Reformatsky reactions, due to their potential to access a broad scope of different heterocycles (Fig. 2; ref. 11).

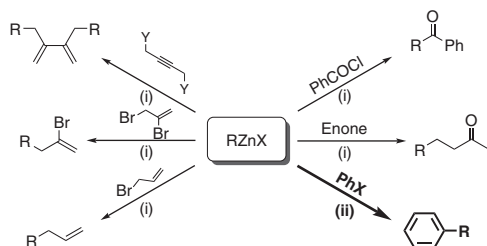


Figure 1 | Reaction of organozinc halides with a variety of electrophiles. These reactions may be done in the presence of (i) Cu(I) or (ii) Pd(0). The Negishi reaction is highlighted in bold. Adapted with permission from ref. 7, Wiley.

Reaction of alkyl and aryl halides with elemental zinc is a typical method for making organozinc reagents¹². Zinc activation processes (e.g., using HCl, DIBAL-H, TMSCl, dibromoethane, or iodine) are exothermic and often require extensive drying before the addition of the organic halide¹³. These metal slurries are highly reactive, air sensitive, and usually pyrophoric in the absence of solvent. In addition, once the organometallic compound is prepared, a decantation or filtration is required to remove unreacted zinc metal and by-product zinc salts in order to obtain a homogeneous solution. Furthermore, reactions with certain substrates, such as aromatic halides, are slow, and harsh conditions are required¹⁴.

Continuous flow techniques can be considered as a step forward to safe and convenient *in situ* generation of organozinc reagents¹⁵. Flow chemistry is typically performed in microreactors with very small internal volumes, so that only small amounts of reagents are mixed at any one time¹⁶. In addition to the small dimensions, the generated products can be immediately used for further reactions in follow-up chemistries¹⁷. Consequently, microreactor technology is perfectly suited for the continuous on-demand production of potentially toxic, reactive, or explosive intermediates, as the hazard of storage and accumulation of large amounts of material is eliminated¹⁸.

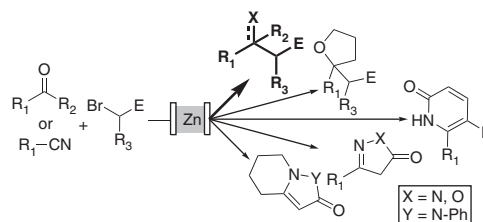


Figure 2 | Range of reactions possible in which alkyl zincbromoacetates are presented using a zinc column. The Reformatsky reaction is highlighted in bold. Adapted with permission from ref. 11, Royal Society of Chemistry.

TABLE 1 | Comparison between batch and flow processes for ethyl zincbromoacetate (**2**) synthesis.

Batch	Flow
Pros • Organozinc slurries are suitable; i.e., the organozinc compound does not need to be completely soluble in the reaction medium • Compatible with poorly soluble reagents Cons • Excess zinc is wasted • Need for decantation and/or filtration to remove excess zinc • Remaining activating reagents and their by-products are present in final solution • Exothermicity is difficult to control	Pros • No zinc waste. Remaining metal reused • Clean solutions of product material are formed; therefore no filtration is required • Control of potential exotherms • Minimal amounts of activators and their by-products are present in final solution • Amount can be prepared on demand and used directly in flow or batch protocols • Collecting the outflow solutions in several reactors allows the running of parallel chemistry Cons • Organozinc concentrations are limited by solubility • Halogenated derivatives must be soluble in THF or dimethylformamide (DMF)

In 2014, we described the continuous formation of organozinc halides using a column filled with metallic zinc and their application in Negishi cross-coupling reactions using a supported catalyst. This work was carried out in collaboration with D.T. McQuade and Z. Miller from Florida State University¹⁹ and presents several advantages over the typical batch procedure^{20–26}. Our two-step process is rapid and high-yielding, and enables the use of a wide range of substrates, avoiding the direct handling of the sensitive organozinc reagents. Another key feature of the flow procedure is that no zinc waste is produced, in contrast to batch protocols, in which the excess of zinc is usually wasted. In this PROCEDURE, the same packed-bed zinc column can be reused in different runs until 25% of the loaded metal remains. Moreover, as only small amounts of reagents are in contact with the zinc at any time, exothermicity of zinc activation and reaction is tightly controlled. These air- and moisture-sensitive compounds are maintained in a closed system under nitrogen atmosphere, which means that they remain stable until they undergo the desired reaction. It is a fast method of obtaining desired intermediates and provides clean solutions with reproducible concentrations in only 4 min of residence time²⁵.

More recently, we illustrated the application of Reformatsky and Blaise reactions for the preparation of a diverse set of valuable intermediates and heterocycles in a one-pot procedure¹¹. The original protocol used in our first report¹⁹ was modified to introduce zinc efficiently into α -bromoacetates. The use of a simplified and greener zinc activation protocol allows us to obtain higher concentrations of desired alkyl zincbromoacetate solutions. The organozinc compounds were reacted with a diverse set of ketones and nitriles to obtain β -hydroxyesters, β -ketoesters and 2,2-disubstituted tetrahydrofuranes. The outflow solutions of these reactions can be collected over the appropriate reagents to obtain a wide range of heterocyclic systems in a batch mode (Fig. 2).

Herein, we describe in detail the setup and synthesis in flow of ethyl zincbromoacetate reagent using a glass column, a metering pump, a suitable column heater and a thermocouple to control the temperature. A comparison of the advantages and disadvantages of the flow protocol with regard to traditional batch chemistry is presented in Table 1. The solution obtained was used in a Reformatsky reaction in flow¹¹ and in a Negishi cross-coupling in batch²⁷ (Fig. 3) following the procedures described in the literature. Details for these two cascade reactions can be found in the Supplementary Methods.

Experimental design

The complete setup for on-column zinc activation and preparation of organozinc reagent **2** (ethyl zincbromoacetate; Fig. 3) is common to all our reported organozinc compounds^{11,19}. The scope of the reaction and conditions for their preparation can be found in Table 2.

For this PROCEDURE, we have chosen an Omnifit SolventPlus column and Vapourtec E Series integrated flow chemistry system. Any other nonmetal column with similar dimensions (Fig. 3) can be used, as long as it can be heated and it is organic solvent-resistant,

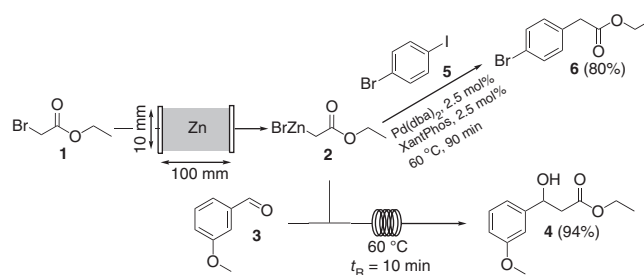
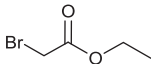
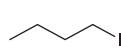
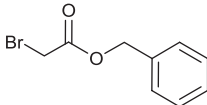
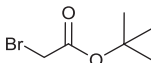
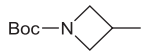
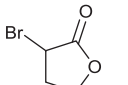
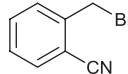
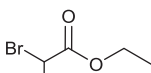
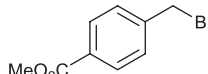
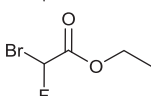
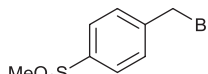
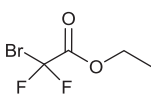
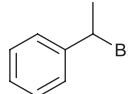
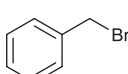
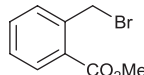


Figure 3 | Preparation of ethyl zincbromoacetate (**2**) in flow and subsequent Reformatsky and Negishi reactions.

TABLE 2 | Scope of organozinc reagents.

Reagent	Solvent (Step 21)	T (°C) (Step 25)	Reagent	Solvent (Step 21)	T (°C) (Step 25)
	THF	40		THF	60
	THF	40		0.5 M LiCl in THF	60
	THF	40		0.5 M LiCl in THF	40
	THF	40		THF	25
	THF	40		THF	25
	THF	40		THF	60
	0.5 M LiCl in THF	100		THF	60
	THF	25		THF	25

All reactions were carried out at 0.5 ml per min flow rate ($t_R = 4$ min).

for instance glass or perfluoroalkoxy alkane (PFA) columns. To reduce the investment cost, the pumping system (Vapourtec E series) can be replaced by syringe or HPLC liquid pumps. As the column reactor (Figs. 4 and 5) is a closed system, column heating can be done by alternative methods, such as a water bath, a heating jacket or a suitable oven.

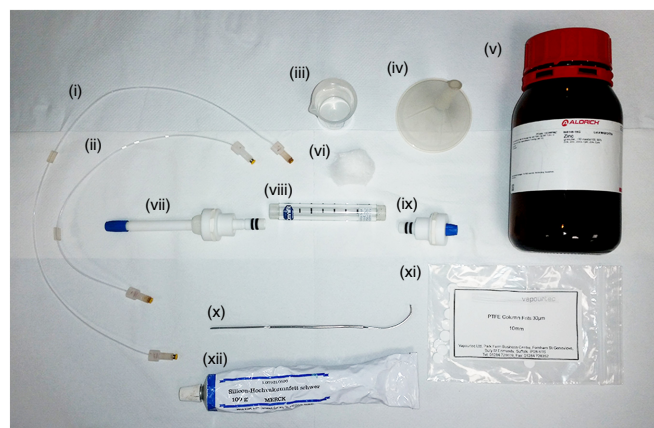


Figure 4 | Equipment for assembly and setup of zinc column reactor. (i) 32-cm outlet PTFE tubing, (ii) 28-cm inlet PTFE tubing, (iii) beaker, (iv) filter funnel, (v) zinc, (vi) cotton, (vii) upper end piece (plunger), (viii) glass tube, (ix) lower end piece, (x) spatula, (xi) frits, and (xii) vacuum grease.

The next steps of the PROCEDURE describe how to perform an example reaction: the synthesis of ethyl zincbromoacetate. In this process, ethyl bromoacetate (**1**) is pumped through the activated zinc column (Steps 22–30; Fig. 3). Using this approach, 50.5 ml of 0.45 M organozinc (**2**) solution was obtained starting from 52.8 ml of 0.47 M ethyl bromoacetate (**1**) solution (91% yield within ~120 min).

A residence time of ~4 min (flow rate = 0.5 ml per min, in our case) for the organic halide **1** within the zinc column is necessary to obtain full conversion. Even though we report the preparation of 50 ml of solution, a standard volume (up to 250–300 ml) for commercially available organozinc reagents can be prepared with a single column. If larger volumes are required, a bigger column can be used, adjusting the flow rate to match a similar residence time. In addition, variation of reaction conditions and parameters such as reaction time, reaction temperature, flow rates and reagent concentration can be easily tuned (Steps 21 and 25), as exemplified in Table 2 for the scope of the reaction.

The organozinc reagent can be used as such in subsequent reactions and concentration can be determined by titration with iodine. The solution can be stored under nitrogen atmosphere at 5 °C. We recommend using the organozinc reagent solution immediately after its production or collecting it into the subsequent reaction vessel. If storage is required, we recommend using it in less than a week due to reagent degradation over time. It is also suggested that the titration be repeated to determine the remaining concentration.

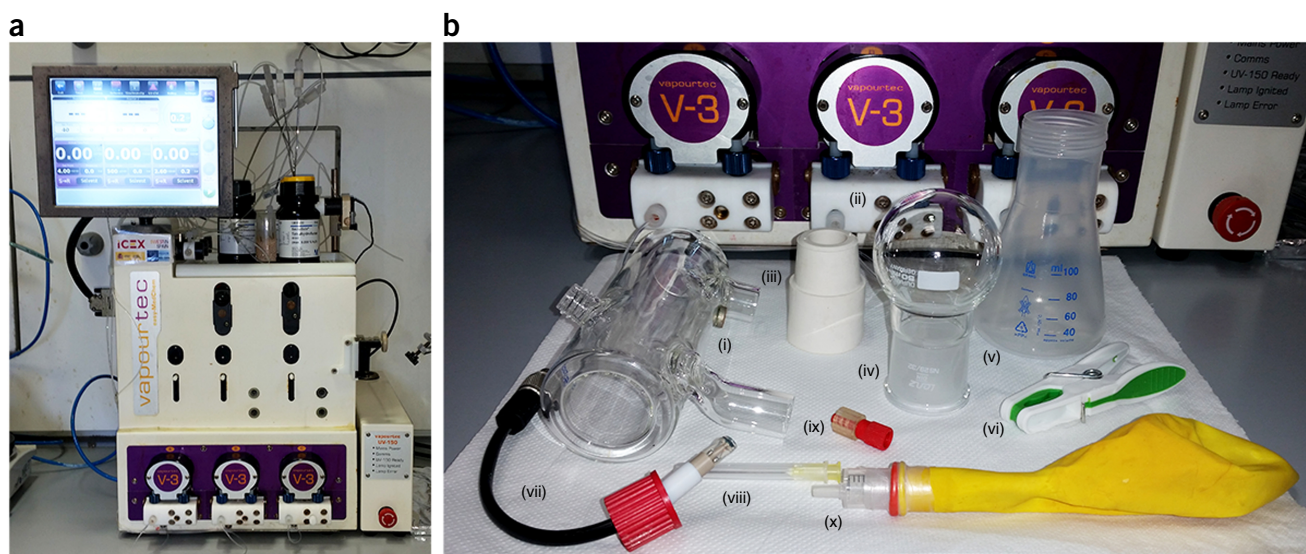


Figure 5 | Materials used for zinc activation. (a) Vapourtec E-Series flow system. (b) Equipment for the zinc activation: (i) glass heat exchanger, (ii) V3 pump blue line, (iii) septum, (iv) round-bottom flask, (v) Erlenmeyer flask, (vi) clothes peg, (vii) temperature sensor, (viii) needle, (ix) needle fitting, (x) balloon.

The aforementioned solution can be mixed with a solution of aldehyde (**3**) in a T-shape mixer and the mixture can be flowed through a 10-ml coil at 60 °C for 10 min to obtain β -hydroxyester (**4**). Alternatively, the organozinc solution can be collected over

a flask containing haloarene (**5**), $\text{Pd}(\text{dba})_2$ and XantPhos. This solution was heated at 60 °C for 90 min to provide ethyl 2-(4-bromophenyl)acetate (**6**). For further details of these two downstream reactions, see the **Supplementary Methods**.

MATERIALS

REAGENTS

! CAUTION All chemicals used in this protocol should be handled with care. Standard protective equipment should be worn (laboratory coat, gloves, safety goggles), and all manipulations should be performed in a ventilated laboratory fume hood.

Reagents for zinc activation

- Zinc (Zn, granular, – 30 mesh $\pm$ 100, 99% (wt/wt); Sigma-Aldrich, cat. no. 565148) **! CAUTION** Zn is self-heating; it may catch fire. Upon contact with water, it releases flammable gases.
- Tetrahydrofuran (THF, dried; SeccoSolv, cat. no. 108107) **! CAUTION** THF is an irritant to the eyes and respiratory system and is highly flammable; it may form explosive peroxides and is suspected of causing cancer.
- Chlorotrimethylsilane (TMSCl, purified by redistillation, $\geq$ 99% (wt/wt); Sigma-Aldrich, cat. no. 386529) **! CAUTION** TMSCl is highly flammable, is harmful upon contact with skin, causes severe skin burns and serious eye damage, is toxic if swallowed or if inhaled and may cause respiratory irritation.
- 1-Bromo-2-chloroethane (98% (wt/wt); Sigma-Aldrich, cat. no. 232750) **! CAUTION** 1-Bromo-2-chloroethane is toxic if swallowed, is harmful upon contact with skin or if inhaled, and may cause respiratory irritation and cancer.

Reagents for the synthesis of ethyl zincbromoacetate

- Ethyl bromoacetate (98% (wt/wt); Sigma-Aldrich, cat. no. 133973) **! CAUTION** Ethyl bromoacetate is flammable and is fatal if swallowed, upon contact with skin or if inhaled.
- THF (dried; SeccoSolv, cat. no. 108107) **! CAUTION** THF is an irritant to the eyes and respiratory system; it is highly flammable, may form explosive peroxides and is suspected of causing cancer.
- **▲ CRITICAL** In the case of certain halogenated reagents, lithium chloride solution can be used instead of pure THF, as it has been reported to increase the stability of the organozinc reagent^{28,29}.
- Lithium chloride solution (LiCl, 0.5 M in anhydrous tetrahydrofuran; Sigma-Aldrich, cat. no. 667544) **! CAUTION** LiCl solution is highly flammable, causes serious eye irritation, may cause respiratory irritation and is suspected of causing cancer.

Reagents for ethyl zincbromoacetate fast titration

- Iodine (I_2 , $\geq$ 98% (wt/wt); Sigma-Aldrich, cat. no. 207772) **! CAUTION** I_2 is harmful upon contact with skin or if inhaled, causes skin and serious eye irritation, may cause respiratory irritation, causes damage to organs (thyroid) through prolonged or repeated exposure if swallowed and is very toxic to aquatic life.
- THF (dried; SeccoSolv, cat. no. 108107) **! CAUTION** THF is an irritant to the eyes and respiratory system; it is highly flammable, may form explosive peroxides and is suspected of causing cancer.

EQUIPMENT

Equipment for assembly and setup of zinc column reactor

- SolventPlus column (bore: 10 mm, length: 100 mm, AF (1 $\times$ adjustable & 1 $\times$ fixed endpiece); Omnifit, cat. no. 006SCC-10-10-AF (http://www.dibaind.com/wp-content/uploads/2015/01/201501_1_OFLW_CAT.pdf)). The needed parts are the glass tube, two 30 μm $\times$ 10 mm polytetrafluoroethylene (PTFE) frits, one fixed end piece as a bed support and one adjustable end piece (plunger) to adjust the bed height.
- PTFE tubing (28 cm and 32 cm $\times$ 1.6-mm o.d., 1.0-mm i.d.; Omnifit, cat. no. 008T16-100-20)
- Ethylene tetrafluoroethylene (ETFE) nuts for 1.6-mm o.d. tubing (1/4-inch-28, natural; Omnifit, cat. no. 002130)
- One-piece PTFE ferrules for 1/4-inch-28 nuts (Omnifit, cat. no. 002310)
- Blanking ETFE plugs for 1/4-inch-28 flat-bottom ports (Omnifit, cat. no. 002320)
- Silicone high-vacuum grease, heavy (Merck, cat. no. 107921)
- Spatula (VWR, cat. no. 82027-528)
- Beaker (25 ml; VWR, cat. no. 89172-038)
- Filter funnel (VWR, cat. no. 30617-150)

Equipment for the zinc activation

- Vapourtec E-Series integrated flow chemistry system (<https://www.vapourtec.com/>)
- V3 tube, organic, type blue (Vapourtec, cat. no. 50-1301)
- Column reactor manifold and thermocouple for 10- and 15-mm-bore columns (Omnifit, cat. no. 50-1013).
- Needle fitting ETFE (male Luer to 1/4-inch-28 female; VWR, cat. no. P-675)

PROTOCOL

- Round-bottom flask (VWR, cat. no. 89090-308)
- Septum (Sigma-Aldrich, cat. no. Z554111)
- Balloon (Sigma-Aldrich, cat. no. Z154997)
- Needle (VWR, cat. no. BD-305175)
- Erlenmeyer flask (100 ml; VWR, cat. no. 24775-002)
- Needle (0.9 × 40 mm; VWR, cat. no. 82007-246)
- Syringe (0.01–1 ml; VWR, cat. no. 720-2561)
- Hypodermic needle (VWR, cat. no. 612-0161)
- Test tube (VWR, cat. no. 10545-922)
- Clothes peg

Equipment for the synthesis of ethyl zincbromoacetate

- Round-bottom flask (2 × 50 ml)
- Septum (2)
- Balloon (2)
- Needle (0.9 × 40 mm) (3)

Equipment for ethyl zincbromoacetate fast titration

- Syringe (0.01–1 ml)
- Hypodermic needle (0.80 × 120 mm)
- Test tube (16 × 100 mm)
- Septum

PROCEDURE

Assembly and setup of zinc column reactor ● TIMING ~20 min, generally

1| Put some vacuum grease on the O-rings of both end pieces (**Fig. 6**).

▲ **CRITICAL STEP** Grease prevents the end pieces from sticking to the glass column.

2| Fit the lower end piece, with its corresponding frit, to one end of the glass column, screwing the retaining nut onto the glass threads. (Optional) For better maintenance of the frit, it is recommended to put a piece of cotton or glass wool between the metal and the frit.

▲ **CRITICAL STEP** Check the frits before assembling the end pieces. They must be white and flexible. If not, clean them with 1 M HCl aqueous solution in an ultrasound bath, then with MeOH and finally with acetone. Dry the frits before putting them back on the end pieces.

3| Weigh ~12 g of zinc in a beaker.

4| Fill the column with zinc using a filter funnel (**Fig. 7a,b**).

5| Assemble the adjustable upper end piece (plunger), with its corresponding frit, screwing the retaining nut onto the glass threads (**Fig. 7c**). (Optional) For better maintenance of the frit, it is recommended to put a piece of cotton or glass wool between the metal and the frit.

6| Compact the zinc by adjusting the plunger with the nut until no empty spaces are observed.

▲ **CRITICAL STEP** Compact gently. If the zinc is too tightly packed, the column may become blocked when the other reagents are added.

7| Insert the column reactor into the glass heat exchanger (**Fig. 8**).

8| Attach the ETFE nut of the 28-cm inlet PTFE tubing to the connection cap of the lower end piece (IN) and tighten it.



Figure 6 | Greasing of the O-rings of the lower end piece in Step 1.

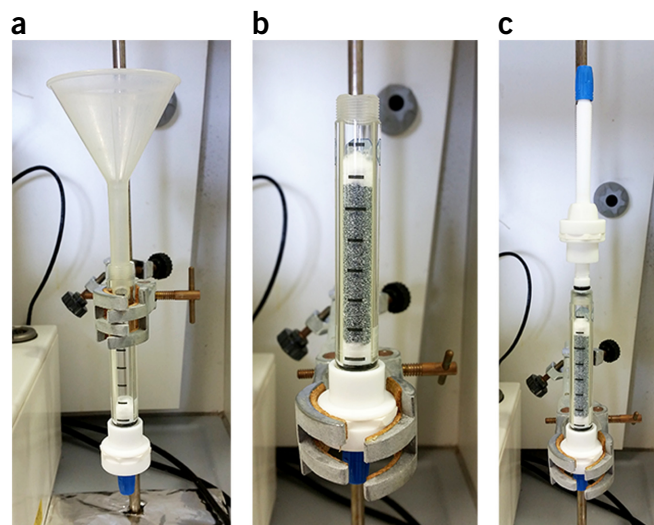


Figure 7 | Preparation of the zinc column. (a) Insertion of cotton at the bottom of the column (Step 2). (b) Filling of the column with zinc and covering it with cotton (Step 4). (c) Assembly of the adjustable upper end piece (plunger) (Step 5).

▲ **CRITICAL STEP** Tighten only finger-tight. If the ferrule is overtightened, leakage can occur. The tubing should not protrude beyond the ferrule, as this could also lead to leakage. Therefore, make sure that the tubing is flush with each ferrule.



Figure 8 | Column reactor inside the glass heat exchanger (Step 7).

9| Attach the ETFE nut of the 32-cm outlet PTFE tubing to the connection cap of the upper end piece (OUT) and tighten it.

10| Attach the needle fitting to the opposite ETFE nut of the 32-cm outlet PTFE tubing and tighten it (**Fig. 9b**).

11| Place the entire setup on the reactor module (**Fig. 9a**).

12| Connect the temperature sensor between the reactor module (temperature sensor connection) and the heat exchanger (**Fig. 9c**).

13| Prime the solvent and reagent lines of a V3 pump with dried THF at 1 ml per min until no more bubbles are observed in the lines (~5 min for each line).

14| Attach the inlet tubing to the V3 pump outlet channel and tighten it (**Fig. 9d**).

15| Using a clothes peg, fasten the end fitting of the outlet tubing to the waste Erlenmeyer flask (**Fig. 9e**). Purge the system by pumping dried THF for 10 min using the V3 pump with a flow rate of 1 ml per min, collecting the discarded liquid into the waste flask. During this process, measure the dead volume of the whole system and the column reactor retention time (t_R). To measure the whole-system dead volume, count the time from when the pump was turned on until the first drop (without air) came out of the system (in our case, $t \approx 6$ min $V \approx 6$ ml). To measure the column reactor t_R , count the time from when the first zinc layer became solvated until the solvent reached the top of the zinc bed (**Fig. 10**) (in our case, $t_R = 2$ min 10 s at a flow rate of 1 ml per min $V = 2.17$ ml). The zinc column setup is now complete and ready to be activated.

▲ **CRITICAL STEP** Check that the entire setup is tightly sealed and that there is no solvent leakage.

? TROUBLESHOOTING

■ **PAUSE POINT** A THF-solvated zinc column can be stored for at least 1 month. However, it is recommended that the system be purged by pumping dried THF for 10 min using the V3 pump with a flow rate of 1 ml per min before zinc activation.

Zinc activation ● **TIMING** ~15 min, generally

16| Prepare 11 ml of 0.6 M TMSCl and 0.24 M 1-bromo-2-chloroethane activating solution in a flask under nitrogen (N_2) atmosphere by dissolving 0.8 ml of TMSCl and 0.2 ml of 1-bromo-2-chloroethane in 10 ml of dried THF.

17| Set the temperature in the Vapourtec system to 40 °C and press the 'RUN' button to heat the reactor column (**Fig. 11a**).

18| Put the reagent line needle into the activating solution (**Fig. 11b**).

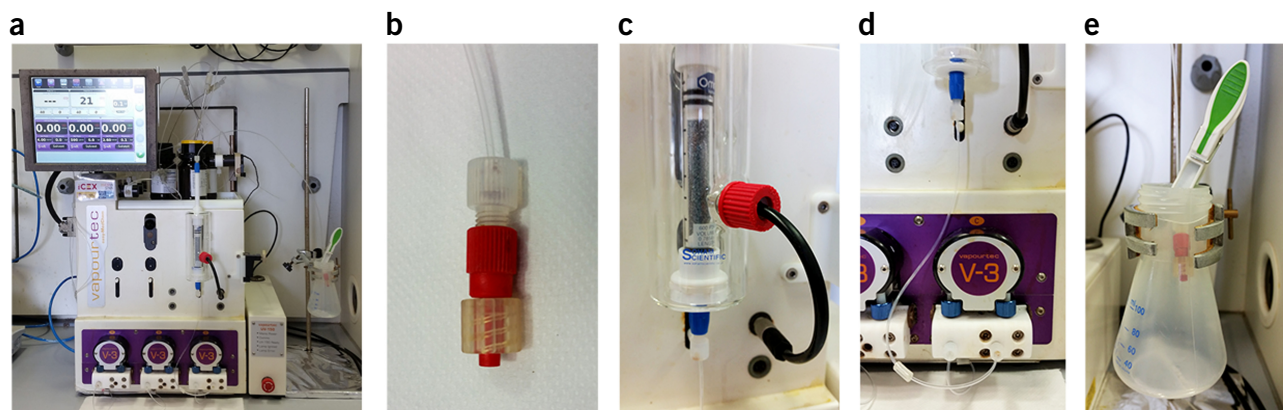


Figure 9 | Setup of the reactor module. (a) The final setup of the reactor module (Step 11). (b) Needle fitting connected to the ETFE nut of the 32-cm outlet PTFE tubing (Step 10). (c) Temperature sensor connected between the reactor module and the heat exchanger (Step 12). (d) Inlet tubing connected to the V3 pump outlet channel (Step 14). (e) Outlet tubing fastened to the waste Erlenmeyer flask (Step 15).

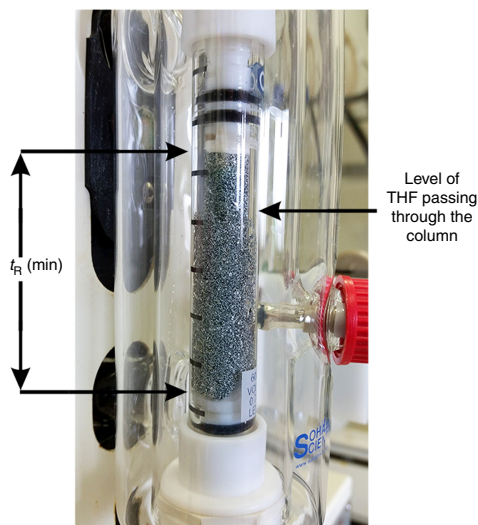


Figure 10 | Measurement of the column reactor residence time (t_R).

▲ **CRITICAL STEP** The needle can become clogged with septum bits when the septum is pierced; if this happens, the needle should be replaced. You should use good-quality septa (i.e., Precision Seal rubber septa).

19 | Pump the activating solution through the zinc column at a flow rate of 1 ml per min. This is done using the reagent line of the V3 pump, and by setting the flow rate at the controller touch screen (**Fig. 11a**).

20 | Stop the pump after 10 min or when extensive bubbling is observed. The zinc column is now ready to be used for organozinc reagent generation.

▲ **CRITICAL STEP** Bubbles of ethylene gas should be observed after ~3 min, indicating that zinc activation is taking place. No temperature variation should be observed. If no bubbles are formed, zinc activation is not complete, and the process must be repeated (check the quality of the reagents and solvent).

Synthesis of ethyl zincbromoacetate (**2**) ● **TIMING** ~120 min, for this specific example

21 | Prepare 52.8 ml of 0.47 M ethyl bromoacetate (**1**) solution in a flask under a N_2 atmosphere, dissolving 4.175 g (25 mmol) of ethyl bromoacetate (**1**) in 50 ml of dried THF.

22 | Replace the activating solution with the ethyl bromoacetate (**1**) solution.

▲ **CRITICAL STEP** The needle can become clogged with septum bits when the septum is pierced; if this happens, the needle should be replaced. You should use good-quality septa (i.e., Precision Seal rubber septa).

23 | Purge a flask with N_2 and cap it immediately; this is to be used for the collection of ethyl zincbromoacetate (**2**) solution.

24 | Attach a needle at the needle fitting of the outlet tubing, and tighten it (**Fig. 12a**).

▲ **CRITICAL STEP** Change the needle after each activation step. The needle becomes oxidized during the process and can contaminate the organozinc solution.

25 | Change the flow rate to 0.5 ml per min and maintain the zinc column temperature at 40 °C. Start to pump the ethyl bromoacetate (**1**) solution through the column (**Fig. 12b**).

26 | Use the whole-system dead volume (~6 ml) to estimate when to start collecting the ethyl zincbromoacetate (**2**) solution. Due to diffusion, dilution occurs at the head

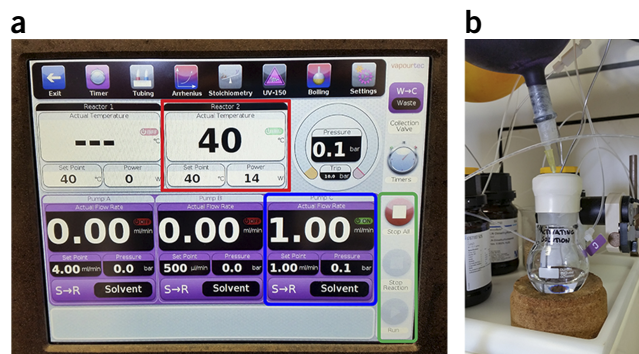


Figure 11 | Reaction conditions and setup for the zinc activation. (a) Reactor module screenshot: (red box) reactor temperature screen; (blue box) real-time flow-rate pump display; (green box) RUN/STOP buttons (Steps 17 and 19). (b) Reagent line needle immersed in the activating solution (Step 18).

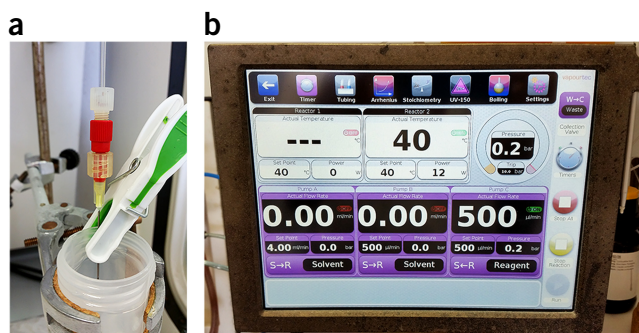


Figure 12 | Setup and conditions for preparation of organozinc reagent. (a) Needle connected to the needle fitting (Step 24). (b) Reactor module screenshot during ethyl zincbromoacetate (**2**) formation (Step 25).

and tail of the solution. To obtain a higher final concentration, it is better to discard some volume of the ethyl zincbromoacetate (**2**) solution (~1 ml of head and ~1 ml of tail solution) and only collect the central part.

? TROUBLESHOOTING

27| To find out when the product is eluting, place ~5 mg of iodine in a test tube and start adding the outlet solution once no more ethylene gas is observed in the tubing. When the color of iodine (purple-brown) vanishes, start collecting the organozinc compound into a flask under a N₂-atmosphere. If the halogenated substrate is an iodide, the organozinc reagent formation can be detected by dropping the outlet solution onto a white tissue and observing the appearance of a yellow spot when the drops become dry. In this setup, we typically start collecting the product at 13 min.

28| After 92 min, switch the pump to solvent (touchscreen) and continue pumping pure dried THF at a flow rate of 0.5 ml per min for a further 10 min (to recover the remaining organozinc reagent **2**) (**Fig. 13**).

▲ **CRITICAL STEP** Collection is stopped after 10 min of flowing solvent, as longer collection will provide larger amounts of solution but lower overall concentrations.

■ **PAUSE POINT** The product solution can be stored under nitrogen atmosphere at 5 °C for 2 weeks. We expect an amount of degradation after the first week. The formation of ethyl zincbromoacetate (**2**) crystals may be observed. If this happens, we advise redissolving the crystals before titration and use of the organozinc reagent.

29| Stop collection and put the out line into the waste flask.

Rinsing and storage of zinc column reactor

● **TIMING** ~10 min, generally

30| Put the reagent line needle into a bottle containing dry THF.

31| Switch the pump to the reagent line and pump dried THF through the system for 10 min at a flow rate of 1 ml per min, to wash the reagent inlet and the zinc column.

32| Stop the pump.

33| Turn off the heating of the column.

34| Storage. To reduce zinc passivation, tighten the blanking ETFE plugs on the connection caps of both end pieces (**Fig. 14**).

■ **PAUSE POINT** The zinc remaining in the column can be used for the next reaction. It should be stored at room temperature (20–25 °C) and is still good after at least 1 month. It is advisable to refill the column when 60–75% of the initial zinc is consumed.

Fast titration of ethyl zincbromoacetate (**2**) solution

● **TIMING** ~5 min, for this specific example

▲ **CRITICAL** The fast titration procedure described here provides an approximation of the solution concentration. For a more accurate value, a proper titration using appropriate analytical instrumentation should be performed.

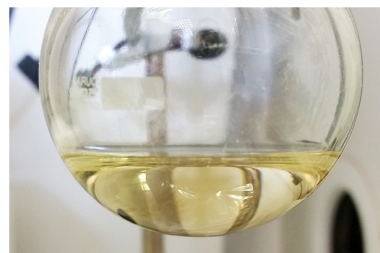


Figure 13 | Clean pale-yellow ethyl zincbromoacetate (**2**) solution.

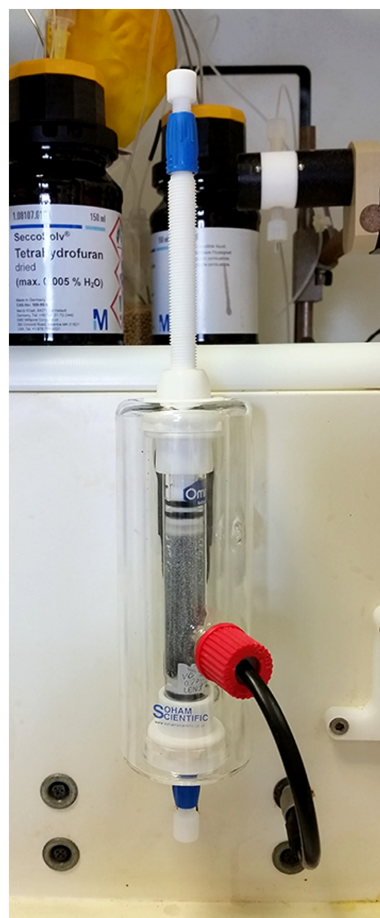


Figure 14 | Zinc column full of dried THF closed by plugs and ready to be stored.

PROTOCOL

35| Weigh ~25 mg of I₂ in a test tube and purge it with N₂. Record the exact amount of I₂.

36| Add 0.2 ml of dried THF.

37| Take 1 ml of ethyl zincbromoacetate (**2**) solution with a 0.01–1 ml syringe.

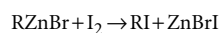
38| Add the organozinc **2** solution dropwise over the I₂ solution (purple-brown) (**Fig. 15a**) until the color vanishes (**Fig. 15b**).

▲ **CRITICAL STEP** In some cases, such as with *tert*-butyl zincbromoacetate, the titration reaction is slow and must be done at 60 °C. If the organozinc solution is colored, the final color of the titration will be the color of the organozinc solution.

? TROUBLESHOOTING

39| Calculate the concentration of the organozinc reagent using the following equations.

General titration equation:



Formula to calculate the organozinc reagent concentration:

$$[\text{RZnBr}](\text{M}) = \frac{\text{W Iodine}(\text{mg})}{\text{MW Iodine} \left(\frac{\text{mg}}{\text{mmol}} \right) * V \text{RZnBr}(\text{mL})}$$

W ≡ weight; MW ≡ molecular weight; V ≡ volume.

? TROUBLESHOOTING

Troubleshooting advice can be found in **Table 3**.

TABLE 3 | Troubleshooting table.

Step	Problem	Possible reason	Solution
15	Leakage	Fittings are either too tight or not tight enough	Check the fittings and either untighten or tighten them
	Blockage or overpressure	Zinc column is overpacked	(i) Unscrew the plunger (ii) Open the column and manually loosen the zinc with a spatula (iii) Empty the column and fill it again
26	Blockage or overpressure	Organozinc reagent has precipitated inside the column or at the outlet	(i) Stop the pumps and purge the system with pure dried THF. If the pressure of the system is normal, start the experiment again (ii) Make sure that the frits are not blocked (iii) Make sure that the needle at the outlet is not blocked
38	Low concentration of product	(i) The reaction was done at too low a temperature (ii) Residence time in the column is too short	(i) Increase the temperature, e.g., to 60 °C (ii) Decrease flow rate, e.g., to 0.25 ml per min

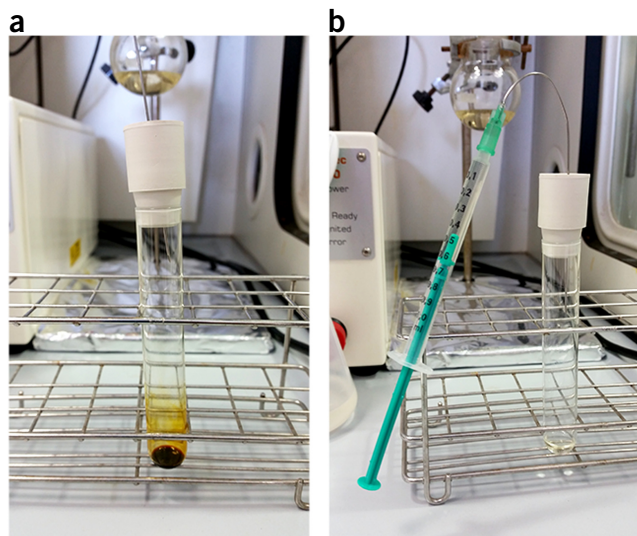


Figure 15 | Color change observed during the fast titration. (a) Dark purple iodine solution in THF before titration (Step 38). (b) Pale yellow solution obtained after titration (Step 38).

TIMING

Steps 1–15, assembly and setup of zinc column reactor: ~20 min, generally

Steps 16–20, zinc activation: ~15 min, generally

Steps 21–29, synthesis of ethyl zincbromoacetate (**2**): ~120 min, for this specific example

Steps 30–34, rinsing and storage of zinc column reactor: ~10 min, generally

Steps 35–39, fast titration of ethyl zincbromoacetate (**2**) solution: ~5 min, for this specific example

ANTICIPATED RESULTS

The typical ethyl zincbromoacetate (**2**) conversion obtained is 91%. In this PROCEDURE, 52.8 ml of 0.47 M ethyl bromoacetate (**1**) solution was passed through the column and 50.5 ml of 0.45 M ethyl zincbromoacetate (**2**) solution was obtained. We must point out that the volume of discarded solution will be always of ~2.0 ml independently of the organozinc reagent volume synthesized. In the following table, we present the system outcomes in a small- and a large-scale production (Table 4). We can observe that the larger the volume of ethyl bromoacetate (**1**) solution that passes through the system, the better the outcome.

TABLE 4 | Comparison between reactions performed on a small and large scale.

	Small scale	Large scale
Volume of the ethyl bromoacetate (1) solution	10.6 ml	52.8 ml
Molarity of ethyl bromoacetate (1)	5 mmol	25 mmol
Reaction time	30 min	115 min
Volume of ethyl zincbromoacetate (2) solution	8 ml	50.5 ml
Concentration of ethyl zincbromoacetate (2) ^a	0.40 M	0.45 M
Yield	64%	91%

^aDetermined by fast titration (Steps 35–39).

Note: Any Supplementary Information and Source Data files are available in the online version of the paper.

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AUTHOR CONTRIBUTIONS M.B. and L.H. carried out the experiments. All authors designed the protocol and J.A. supervised the project. All authors contributed to the writing of the manuscript.

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